

Updated Instructions for App Designer/Coder

Project: SOET Attendance, Mentorship & Faculty Leave-Arrangement Monitoring App – SGT University

1. Core Objective

Develop a role-based app for SOET, SGT University, where teachers mark attendance only for subjects assigned to them, mentors monitor only their assigned mentees, HoDs approve leave/attendance exceptions, and the Dean monitors overall attendance, mentorship, faculty compliance, and detention-risk reports.

2. User Roles

The app shall have the following logins:

1. Teacher Login
2. Mentor Login — may be part of Teacher Login
3. Activity Coordinator Login
4. HoD Login
5. Dean Login
6. Admin Login

3. Teacher Login Requirements

A. Subject Attendance

The teacher shall see **only the subjects officially assigned to him/her** by Admin/HoD.

The teacher should not manually search all subjects. The dashboard should display only assigned subjects with:

Field	Details
Programme	Auto-mapped
Semester	Auto-mapped
Section	Auto-mapped
Subject Name	Auto-mapped
Subject Code	Auto-mapped
Lecture Slot	As per timetable
Date	Auto/current date with edit permission as per rule

The teacher will select the relevant assigned subject/slot and mark:

- Present

- Absent
- No Class Conducted
- Class Taken by Arrangement Teacher

Attendance should be locked after a defined time, e.g., 24 hours, unless reopened by HoD/Admin.

B. Mentee List Visibility

Each teacher shall see **only the mentees assigned to him/her**.

The mentees may belong to:

- Same batch
- Different batches
- Different programmes/sections, if assigned by Admin/HoD

The mentor dashboard shall display a mentee-wise table with subject headers and attendance percentage.

Suggested Mentee Attendance Table

Student Name	Roll No.	Batch	Subject 1	Subject 2	Subject 3	Subject 4	Overall %	Risk Status
Student A	101	B.Tech Sem III	CSE 82%	76%	68%	91%	79%	Safe
Student B	102	B.Tech Sem III	AI 71%	64%	58%	80%	68%	At Risk

Colour Coding

Use colour coding for quick monitoring:

Attendance Range	Colour	Status
85% and above	Green	Good
75% to 84%	Light Green/Blue	Safe
65% to 74%	Yellow/Orange	Warning
Below 65%	Red	Critical
Approved leave/activity pending	Grey/Amber	Pending Approval

Mentor should be able to click on any subject percentage to see:

- Total classes held
- Classes attended
- Approved activity/leave attendance
- Absences

- Pending regularization
- Remarks/counselling history

4. Mentor Attendance Regularization

Mentor can initiate attendance regularization only for assigned mentees under approved categories:

- Sick leave
- Internship
- Project work
- University duty
- Department-approved academic activity
- Approved external event
- Placement/training activity

Workflow:

Mentor → Submit request with proof/remarks → HoD Approval → Attendance updated

The system must maintain:

- Student name
- Reason
- Date/slot
- Supporting document
- Mentor remarks
- HoD approval/rejection
- Audit trail

5. Faculty Leave and Class Arrangement Module

A separate module shall be provided in Teacher Login for faculty leave and class arrangement.

A. Leave Application by Teacher

When a teacher is applying for leave, the teacher must make class arrangements before submitting leave.

The teacher shall select:

- Leave date
- Leave type
- Reason for leave
- Assigned lectures affected
- Subject/section affected

- Proposed arrangement teacher for each lecture

B. Arrangement Teacher Selection

For each affected lecture, the system shall show a **pre-fed list of available teachers** who are free during that lecture slot.

The system should automatically filter teachers based on:

- Free timetable slot
- Department/faculty availability
- Teaching load
- Eligibility/competence mapping, if available
- No clash with existing lecture
- Not already assigned for another arrangement during same slot

The leave-applying teacher can select a suitable arrangement teacher for each lecture.

Suggested Table

Date	Slot	Subject	Section	Original Teacher	Available Arrangement Teachers	Selected Teacher	Status
12 Aug	10:00–11:00	Data Structures	CSE III	Dr. X	Dr. A, Dr. B, Dr. C	Dr. B	Pending Acceptance
12 Aug	12:00–01:00	AI Basics	AI III	Dr. X	Dr. D, Dr. E	Dr. D	Pending Acceptance

C. Approval by Arrangement Teacher

The selected arrangement teacher must approve/accept the class arrangement.

Workflow:

Leave Applying Teacher → Select Arrangement Teacher → Arrangement Teacher Accepts/Rejects → HoD Reviews Leave

If the arrangement teacher rejects, the original teacher must select another available teacher.

D. HoD Leave Approval

The HoD shall see the leave application along with class arrangement status.

HoD dashboard should show:

- Leave applicant name
- Leave date
- Affected lectures
- Arrangement teacher selected
- Arrangement teacher acceptance status
- Remarks
- Final approval/rejection option

HoD should approve leave only after arrangement status is complete, unless exempted.

6. Arrangement Class Attendance Rule

Arrangement classes must be treated carefully in the system.

Key Rule

Arrangement classes shall be counted for student attendance but shall not be counted as class completion by the original subject teacher.

This means:

Parameter	Treatment
Student attendance	Counted
Detention calculation	Counted
Subject attendance percentage	Counted
Class completion schedule of original teacher	Not counted
Workload completion of original teacher	Not counted
Arrangement teacher record	Counted as arrangement duty
HoD/Dean monitoring	Visible separately

This ensures that students are not detained due to faculty leave if they attended the arranged class.

7. Attendance Marking for Arrangement Classes

Once the arrangement teacher accepts the class:

- The arranged lecture appears in the arrangement teacher's dashboard.
- The arrangement teacher can mark attendance for that class.
- Attendance will be reflected in the student's subject attendance.
- The class will be tagged as **Arrangement Class**.
- It will not increase the original teacher's completed class count.

Suggested attendance tag:

- Regular Class
- Arrangement Class
- Activity Attendance
- Approved Leave/Internship/Project
- No Class Conducted

8. Detention List Rule

While generating the detained/shortage list:

The system shall count the following as valid attendance:

- Regular class attendance
- Arrangement class attendance
- Approved sick leave, if allowed under university policy
- Approved internship/project attendance
- Approved activity attendance
- Approved university duty

Arrangement class attendance must be included in student attendance percentage and detention calculation.

However, arrangement classes must be separately visible in reports so that class completion and faculty workload are not incorrectly inflated.

9. Activity-Based Attendance

For seminars, workshops, industrial visits, project reviews, placement activities, training programmes, etc.:

- Activity Coordinator marks student participation.
- HoD approves the activity attendance.
- Subject teacher marks “No Class Conducted due to Approved Activity” where applicable.
- Students receive attendance for the approved activity slot.
- No duplicate attendance shall be allowed for the same student and same slot.

10. HoD Dashboard Requirements

HoD shall be able to view:

1. Subject-wise attendance
2. Teacher-wise attendance marking status
3. Mentor-wise mentee attendance

4. Low attendance students
5. Pending mentor regularization requests
6. Pending activity attendance approvals
7. Faculty leave applications
8. Class arrangement status
9. Arrangement teacher acceptance/rejection
10. Regular class vs arrangement class report
11. No Class Conducted report
12. Detention-risk list

11. Dean Dashboard Requirements

Dean shall have school-level monitoring access across all SOET departments.

Dean should be able to view:

- Department-wise attendance summary
- Programme-wise and semester-wise attendance
- Students below 75%
- Mentor intervention status
- Faculty attendance marking compliance
- Faculty leave and arrangement class summary
- Arrangement classes conducted
- Activity-based attendance
- No Class Conducted cases
- Detention list with detailed attendance breakup
- Monthly trend report

Dean should mainly have monitoring rights, with limited administrative override if permitted.

12. Admin Configuration

Admin shall configure:

- Faculty master
- Student master
- Programme/semester/section
- Subject allocation
- Teacher-subject mapping
- Mentor-mentee mapping
- Timetable
- Lecture slots
- Free teacher availability
- HoD mapping
- Dean mapping
- Leave types

- Attendance rules
- Detention threshold
- Activity categories
- Approval hierarchy

13. Essential Business Rules

1. Teacher shall see only the subjects assigned to him/her.
2. Teacher shall mark attendance only for assigned subjects or approved arrangement classes.
3. Mentor shall see only assigned mentees.
4. Mentee attendance table shall show subject-wise attendance with colour coding.
5. Leave application must include class arrangement details.
6. Arrangement teacher must accept the class before HoD leave approval.
7. HoD shall approve leave after checking arrangement status.
8. Arrangement class attendance shall count for student attendance and detention calculation.
9. Arrangement class shall not count in original teacher's class completion schedule.
10. Activity attendance shall require coordinator entry and HoD approval.
11. No duplicate attendance shall be allowed for the same student, date, and slot.
12. All approvals, changes, and attendance edits must have audit trail.
13. Reports must be downloadable in Excel and PDF.

Updated Instruction: Extra Class / Course Completion Provision

Add the following module to the earlier app instructions.

14. Extra Class / Make-Up Class Module

The app shall include a provision for **extra classes/make-up classes** in case a course is not completed due to:

- Faculty leave
- Holidays
- Events
- Academic activity clashes
- No Class Conducted entries
- Slow course progress
- Any other valid academic reason

This module will allow the subject teacher to compensate pending lectures and complete the syllabus/course plan.

A. Extra Class Selection by Teacher

The teacher shall see only the batches/sections and subjects assigned to him/her.

For scheduling an extra class, the teacher shall select:

Field	Requirement
Subject	From assigned subjects only
Programme/Semester/Section	Auto-mapped from subject allocation
Date	Selected by teacher
Available Slot	From pre-fed dropdown based on free slots of that student batch
Class Type	Extra Class / Make-Up Class
Reason	Leave, holiday, no class conducted, syllabus backlog, etc.
Topic Covered	Mandatory entry
Remarks	Optional

B. Available Slot Dropdown

The system shall show only those slots where the concerned student batch/section is free.

The available slots should be generated from:

- Student batch timetable
- Free lecture periods
- Vacant slots
- Lunch/break rules, if applicable
- Working day calendar
- Special timetable restrictions
- HoD/Admin-approved extra class slots

The system should prevent clashes with:

- Existing regular classes
- Approved activities
- Industrial visits
- Exams/tests
- Other already scheduled extra classes
- Faculty's own timetable clash

C. Attendance for Extra Class

Once the extra class is scheduled and conducted, the teacher shall mark attendance for that class.

This attendance shall be:

Parameter	Treatment
Student attendance	Counted
Detention calculation	Counted
Subject attendance percentage	Counted
Teacher's course completion lecture count	Counted
Syllabus/course progress	Counted
Replacement for No Class Conducted lecture	Allowed
Audit trail	Mandatory

D. Difference Between Arrangement Class and Extra Class

The system must clearly distinguish between **Arrangement Class** and **Extra Class**.

Feature	Arrangement Class	Extra Class / Make-Up Class
Reason	Teacher is on leave/unavailable	
Taken by	Arrangement teacher	
Attendance counted for student	Yes	
Counted in detention calculation	Yes	
Counted in original teacher's course completion	No	
Counted in arrangement teacher duty record	Yes	
Purpose	Avoid student attendance loss during teacher absence	
Feature	Extra Class / Make-Up Class	
Reason	Course backlog/syllabus completion/no class conducted compensation	
Taken by	Original subject teacher	
Attendance counted for student	Yes	
Counted in detention calculation	Yes	
Counted in original teacher's course completion	Yes	
Counted in syllabus/course progress	Yes	
Purpose	Complete pending syllabus and lecture requirement	

E. Treatment of "No Class Conducted"

If a regular lecture is marked as **No Class Conducted**, it shall not be counted as:

- Class held
- Student attendance opportunity

- Teacher's completed lecture
- Course completion progress

However, the teacher may later compensate it by scheduling an **Extra Class/Make-Up Class** through available batch slots.

Once the extra class is conducted:

- It shall count in student attendance.
- It shall count in detention calculation.
- It shall count in teacher's total lectures for course completion.
- It shall be visible separately as an extra/make-up lecture.

F. HoD Monitoring for Extra Classes

HoD dashboard shall show:

- Subject-wise regular classes conducted
- No Class Conducted lectures
- Extra classes scheduled
- Extra classes completed
- Attendance in extra classes
- Teacher-wise course completion status
- Syllabus progress
- Pending lecture backlog
- Extra class approval status, if approval is enabled

HoD may approve or review extra classes depending on university rules.

G. Dean Monitoring for Extra Classes

Dean dashboard shall include:

- Department-wise course completion status
- Teacher-wise lecture completion
- Subject-wise backlog
- No Class Conducted vs Extra Class compensation
- Extra class attendance impact on detention list
- Monthly academic compliance report

Updated Key Business Rules

1. Teacher shall see and select only the subjects assigned to him/her.
2. Teacher shall see only mentees assigned to him/her.

3. Mentee table shall show subject-wise attendance with colour coding.
4. Arrangement classes shall count for student attendance and detention but not for original teacher's course completion.
5. Extra classes/make-up classes taken by the original teacher shall count for student attendance, detention calculation, and teacher's course completion.
6. No Class Conducted lectures shall not count in course completion or student attendance.
7. No Class Conducted lectures may be compensated through extra classes using available batch slots.
8. Extra class slots shall be selected from pre-fed dropdowns based on batch availability.
9. The system shall prevent timetable clashes for students and teachers.
10. All regular, arrangement, activity, leave, and extra class attendance must be separately tagged in reports.
11. Detention list shall include valid attendance from regular classes, arrangement classes, approved activities, approved regularization, and extra classes.
12. Course completion report shall exclude No Class Conducted and arrangement classes but include regular and extra classes taken by the concerned subject teacher.

Updated One-Line Requirement for Developer

Build a role-based SOET attendance, mentorship, leave-arrangement, and course-completion monitoring app where teachers see only assigned subjects and mentees, mentors monitor colour-coded mentee attendance, faculty leave requires accepted class arrangements, arrangement classes count for student attendance but not teacher course completion, and extra/make-up classes selected from available batch slots count for both student attendance/detention and teacher course completion, excluding No Class Conducted lectures.